ASK PYTHON DOCS

The Librarian | Project report

Objective and corpus

The Librarian helps students answer Python questions with cited evidence from official Python 3.12 documentation: 73 HTML pages, 784 sections and 2,632 chunks. It preserves section titles, version and source URLs. Chunks target 220 words with up to 40 words of overlap; this matched the larger 320-word configuration's development recall with shorter contexts.

Evaluation and results

The 120-question dataset contains 55 answerable development cases, 55 answerable test paraphrases and 10 out-of-scope test cases. Symbol-based relevance labels require human review. The historical retrieval results below predate Chroma and must be rerun for the current architecture.

Retrieval method	Recall@5	MRR@5	p95 (ms)
BM25	0.945	0.852	9.0
Dense vector	0.909	0.772	19.8
Hybrid fusion	0.964	0.842	25.9
Hybrid + reranker	0.945	0.874	113.8

Optimization

In the earlier Groq benchmark, reducing reranking from 24 to 8 candidates lowered end-to-end p95 from 6,930 to 5,452 ms (21.3%); the correctness proxy rose from 0.846 to 0.877. The proxy checks expected symbols and cited passages, not human semantic correctness. These results do not establish current OpenAI/Chroma performance; the recorded optimization target was not met.

Limitations and next steps

Tables, nested definitions and long code can lose structure or exceed embedding limits. Citation checks do not prove semantic support; abstention uses a heuristic threshold. Shared development/test topics limit generalization claims. Next steps: human relevance and answer review, new Chroma/OpenAI benchmarks, user feedback and concurrent-load testing. Public deployment and user validation remain pending. Token usage is recorded; OpenAI cost requires configured prices.

Source: docs.python.org/3.12/ | Project design.md and recorded results/ experiments.

OUR ARCHITECTURE DESIGN

The Librarian | System pipeline

Architecture design

BeautifulSoup extracts sections and chunks. Local all-MiniLM-L6-v2 embeddings feed persistent Chroma; BM25 supplies keyword matches. Reciprocal rank fusion combines both rankings, then ms-marco-MiniLM-L-6-v2 reranks 8 candidates (optimized) or 24 (baseline) and returns three passages. OpenAI defaults to gpt-4.1-mini; automatic selection prefers OpenAI, then Groq, then the extractive offline baseline. The app validates citation IDs and OpenAI supporting quotations, and can abstain when evidence is insufficient. FastAPI serves the browser UI and JSON API, with a bounded exact-query cache and serialized inference. Docker packaging is included.

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